Omar Tronelli

omar.tronelli@gmail.com +39 392 305 5676 LinkedIn Portfolio

PROFILE

MSc graduate in quantitative finance (110/110 *cum laude*) with hands-on work in Python on derivatives pricing and credit: CDS and counterparty risk (CVA, netting), option Greeks and delta-hedged P&L, yield curves and default-probability models.

EDUCATION

Sapienza University of Rome Rome, Italy

MSc Financial Risk and Data Analysis | Final grade average: 29/30 Sep 2024 – Oct 2026

Thesis: "Counterparty Risk in a CDS Portfolio: CVA, Netting and Contagion"
Supervisor: Prof. Claudia Ceci
Coursework: Quantitative Financial Modeling, Probability & Stochastic Processes, Machine Learning, Risk Management & Capital Requirements

BSc Economics and Finance Sep 2021 – Oct 2024

Thesis: "Predictive Models for Exchange Rates"
Supervisor: Prof. Marco Di Pietro
Coursework: Econometrics, Advanced Statistics, Monetary Policy, Financial Intermediaries

WORK EXPERIENCE

Gaussian Process Model Selection Mar 2025 – Dec 2025

- Research collaboration with two senior researchers, Dr Francesca Panero and Dr Sahoko Ishida; in charge of the code, from implementation to weekly walkthroughs.
- Built an object-oriented Python package of about 1,200 lines (custom additive/ANOVA kernel classes in GPflow, experiment pipeline); ran batch jobs on Linux HPC (SLURM).

ACADEMIC PROJECTS

MSc Thesis: Counterparty Risk in a CDS Portfolio

- Modeled a CDS portfolio facing a defaultable counterparty, with contagion between CIR default intensities; derived prices and netted CVA as recursive systems of Cauchy PDEs.
- Solved them by finite differences in Python, validated vs closed forms and Monte Carlo. In the cases run, contagion raised CVA (+150% for a two-name same-sign portfolio) and moved the netting benefit in opposite directions: 26.6% → 7.5% same-sign, 28.5% → 33.5% opposite-sign (illustrative parameters).

Options, Greeks and Delta-hedged P&L

- Derived Black–Scholes–Merton by replication and risk-neutral valuation, and analyzed the Greeks.
- Derived the Gamma–Theta Break-even Range (GTBR): the daily move at which the gamma P&L of a delta-hedged option offsets its theta ($\approx \sigma/\sqrt{252}$ at the money).
- Backtested a SPY breakout strategy on the GTBR (VIX-based threshold) with stops, costs, risk-based sizing and bootstrap tests, on two years of hourly and thirty years of daily data: no statistically significant edge on the data tested.

Credit Risk Model: Probability of Default and Vasicek Portfolio

- Built an end-to-end default-probability model in Python on about 5 million Freddie Mac loan-years: logit/probit, random-forest benchmark, out-of-time validation (AUC, Brier score, calibration).
- Implemented a Vasicek single-factor portfolio model with Basel-style worst-case PD (99.9% worst-case PD about 11 times the unconditional PD).

SOFR OIS Yield Curve Construction

- Merged 86 Excel files of SOFR swap rates into one cleaned panel and bootstrapped the OIS zero curve in Python, with rateslib and with a hand-written bisection solver.

SKILLS & INTERESTS

Technical Skills

Programming: Python (NumPy, Pandas, TensorFlow, scikit-learn, GPflow), R, C (basic)
Tools: Git/GitHub, HPC with SLURM, CLI-based AI coding agents

Leadership, Teamwork & Personal Skills

Detail-oriented and accurate, with a strong sense of responsibility and inquisitiveness; able to lead and collaborate within teams, and to work well under pressure on tight deadlines while maintaining precision.

Languages: Italian (native), English (fluent, C1), Spanish (B2), Portuguese (basic)

Interests: Financial markets, Programming, Free-diving, Hiking, Chess